Climate Action Assistant Chatbot

1. Selected SDG and Reason for Selection

Selected SDG

SDG 13 – Climate Action

What the SDG Focuses On

SDG 13 focuses on taking urgent action to combat climate change and its impacts. It promotes environmental awareness, sustainable practices, and the reduction of greenhouse gas emissions.

Why This Issue Is Important

Climate change is one of the biggest global challenges today. Rising temperatures, pollution, deforestation, and excessive use of natural resources negatively affect the environment and human health. Creating awareness about climate action can help individuals adopt sustainable habits and contribute to environmental protection.

2. Problem Statement

Many people are unaware of how their daily activities contribute to climate change. Excessive electricity consumption, plastic waste, water wastage, and environmental pollution continue to increase due to a lack of awareness and proper guidance.

Where Does This Problem Exist?

This problem exists globally in homes, schools, colleges, workplaces, and public spaces.

Who Is Affected?

- Students
- Families
- Communities
- Future generations

Why Is It Serious?

Climate change causes environmental degradation, biodiversity loss, water shortages, pollution, and extreme weather conditions.

What Could Happen If It Is Not Solved?

Environmental damage will continue to increase, affecting natural resources, public health, and future sustainability.

3. Project Objective

The objective of this project is to develop an AI-powered chatbot that educates users about climate change and encourages environmentally friendly practices. The chatbot provides guidance on energy conservation, water conservation, plastic waste reduction, and environmental protection.

4. Proposed Solution

Project Name

Climate Action Assistant Chatbot

What Is The Project?

The Climate Action Assistant Chatbot is an AI-powered chatbot developed using Botpress. It helps users learn about climate change and provides suggestions for adopting sustainable practices.

How The System Works

Users interact with the chatbot by asking questions related to climate change, electricity saving, water conservation, plastic waste reduction, and environmental protection. The chatbot processes user queries and provides instant responses based on predefined instructions.

How It Helps Solve The Problem

The chatbot spreads awareness about environmental issues and encourages users to adopt sustainable habits in their daily lives.

Who Will Use The Application?

- Students
- Teachers
- Environmental enthusiasts
- General public

5. Project Features

Climate Change Awareness

Provides information about climate change and its environmental impact.

Energy Saving Tips

Suggests practical ways to reduce electricity consumption and save energy.

Plastic Waste Reduction Guidance

Promotes the use of reusable products and reduction of plastic waste.

Water Conservation Tips

Provides recommendations for saving water and preventing wastage.

Environmental Protection Awareness

Educates users about eco-friendly habits and sustainability.

Fallback Response

Handles unrelated questions and redirects users toward climate action topics.

6. Technology Stack

- Botpress
- Artificial Intelligence (AI)
- Natural Language Processing (NLP)
- GitHub

7. Project Screenshots

Figure 1: Bot Dashboard/Home Page

This screenshot shows the home page of the Climate Action Assistant Chatbot developed using Botpress. It displays the chatbot instructions, knowledge base configuration, and development environment.

Figure 2: Workflow Diagram

This screenshot shows the workflow of the chatbot consisting of a Start node, Autonomous Node, and End node. User queries are processed through the Autonomous Node to generate appropriate responses.

Figure 3: Autonomous Node Instructions

This screenshot displays the instructions provided to the chatbot. These instructions define the chatbot's behaviour and responses related to climate change, energy conservation, water conservation, plastic waste reduction, and environmental protection.

Figure 4: Chatbot User Interactions

This screenshot demonstrates multiple chatbot interactions. The chatbot successfully answers questions related to climate change, electricity saving, plastic waste reduction, water conservation, and tree plantation. It also provides a fallback response for unrelated queries.

Figure 5: Deployed Chatbot Interface

This screenshot shows the publicly deployed Climate Action Assistant Chatbot. Users can access the chatbot through a web-based interface and interact with it to learn about climate action and sustainable practices.

8. Open Source Repository and Project Link

GitHub Repository

https://github.com/shloka0802/Climate-Action-Assistant-Chatbot

Project Demo Link

https://cdn.botpress.cloud/webchat/v3.6/shareable.html?configUrl=https://files.bpcontent.cloud/2026/06/14/18/20260614185314-FCHVT9R1.json

9. Future Scope

- Voice-enabled chatbot interaction
- Mobile application integration
- Multi-language support
- Carbon footprint tracking

- Personalized environmental recommendations
- Real-time environmental awareness updates

10. Conclusion

The Climate Action Assistant Chatbot was successfully developed using Botpress to support SDG 13: Climate Action. The chatbot provides users with information about climate change, energy conservation, water conservation, plastic waste reduction, and environmental protection. It promotes environmental awareness and encourages sustainable practices. This project demonstrates how Artificial Intelligence can be used to educate people and contribute to solving real-world environmental challenges.